Table S1. Prior distributions used in the geospatial Bayesian leaf wax model

Parameter	**Description**	**Prior**	**Informativeness**	**Justification**
β_0	Global intercept for $\delta^2\mathrm{H}_{\mathrm{wax}}$	normal(0, 5)	Weakly informative	Wide SD allows intercept to adapt to data; centered at 0 for standardized $\delta^2\mathrm{H}_{\mathrm{wax}}$
β_{OIPC}	Global coefficient for $\delta^2\mathrm{H}_{\mathrm{precip}}$	normal(0.8, 0.3)	Informative	Prior based on OLS; SD of 0.3 allows 0.2–1.4 range
λ_{decay}	Exponential decay for spatial scale weighting	lognormal(2.5, 0.5)	Weakly informative	Centers at ∼12 km based on preliminary analysis; allows flexibility
β_{C4}	C_4 fraction coefficient	normal(0, 2)	Weakly informative	Centered at 0 with wide SD
β_{tree}	Tree fraction coefficient	normal(0, 2)	Weakly informative	Centered at 0 with wide SD
β_{shrub}	Shrub fraction coefficient	normal(0, 2)	Weakly informative	Centered at 0 with wide SD
β_{grass}	Grass fraction coefficient	normal(0, 2)	Weakly informative	Centered at 0 with wide SD
β_{precip}	Precipitation coefficient	normal(0, 0.02)	Informative	Small expected effect based on physical reasoning
$\beta_{\mathrm{elev}}[k]$	B-spline coefficients	$\beta_{\mathrm{elev}}[1] \sim \mathrm{Normal}(0,\ \tau_{\mathrm{elev}})$; $\beta_{\mathrm{elev}}[k] \sim \mathrm{Normal}(\beta_{\mathrm{elev}}[k-1],\ \tau_{\mathrm{elev}})$ for $k > 1$	Moderately informative	Encourages smooth elevation response
τ_{elev}	Elevation spline smoothness	normal(0, 1)	Moderately informative	Controls smoothness between knots
$\sigma_{\mathrm{intercept}}$	SD of spatial intercept GP	exponential($\lambda_{\mathrm{intercept}}$) where $\lambda_{\mathrm{intercept}} = -\log(0.05)/20$	PC prior	$\mathrm{P}(\sigma > u) = \alpha$; with $u = 20‰$, $\alpha = 0.05$
σ_{slope}	SD of spatial slope GP	exponential(λ_{slope}) where $\lambda_{\mathrm{slope}} = -\log(0.05)/0.3$	PC prior	$\mathrm{P}(\sigma > u) = \alpha$; with $u = 0.3$, $\alpha = 0.05$
$\log(\rho_{\mathrm{spatial}})$	Log length scale for GP	normal(−1.0, 0.4)	Moderately informative	Centered on ∼1850 km scale for regional climate variations
$z_{\mathrm{intercept}}[k]$, $z_{\mathrm{slope}}[k]$	Standardized spatial effects at knots	normal(0, τ_k); $\tau \in [0.5, 0.8] \times$ range_factor (slopes only)	Density/range-adaptive	Adaptive regularization to data density and OIPC range; prevents unrealistic values
σ	Residual SD (includes nugget)	normal(0, 2)	Weakly informative	Determined by data
$\beta_{\mathrm{OIPC}\times\mathrm{C4}}$	OIPC × C_4 interaction	normal(0, 0.5)	Moderately informative	Moderate interactions expected
$\beta_{\mathrm{OIPC}\times\mathrm{PFT}}$	OIPC × PFT interactions	normal(0, 0.5)	Moderately informative	Moderate interactions expected